

ISIC competition Deep Learning MHealth

kaggle.com/competitions/isic-competition-deep-learning-m-health/data



Data Description

Welcome to the **2025 Kaggle Competition of AI Applied to Medicine at UC3M!** Here's a quick guide to the provided data files and their structure. This page should help you answer:

- Which files do I need?
- Which columns or features are relevant to prediction?
- What is the correct submission format?

Files

1. `train-image.hdf5`

- A single HDF5 file containing the **training images**.
- Each image is stored under a key corresponding to its `isic_id`.
- You can decode each image (in RGB) using OpenCV or other libraries.

2. `test-image.hdf5`

- A single HDF5 file containing the **test images**.
- Similar to the training file, but for images that you'll generate predictions on.

3. `train-metadata.csv`

- Metadata for the training set.
- Includes columns such as:
 - `isic_id`: Unique image identifier (matches keys in `train-image.hdf5`).
 - `target`: Binary label (0 = benign, 1 = malignant).
 - Additional clinical and image-based features (e.g., `age_approx`, `sex`, `anatom_site_general`).

4. `test-metadata.csv`

- Metadata for the test set.
- Similar structure to `train-metadata.csv` **without** the `target` column.

5. `sample_submission.csv`

- A sample submission file **in the correct format**.
- Two columns: `isic_id` and a placeholder `target` value (e.g., 0.0).
- Replace the placeholder values with your predicted probabilities (range [0, 1]) for each test image.

Prediction Task

Your objective is to predict the probability that a given lesion (`isic_id`) is **malignant**. You must provide these probabilities for **all** test images in `test-image.hdf5`, indexed by `test-metadata.csv`.

Columns in the Metadata (Selected)

`train-metadata.csv`

- `isic_id`
Unique ID (e.g. "ISIC_0015670"). Matches the dataset key in `train-image.hdf5`.
- `target`
Binary label: 0 (benign) or 1 (malignant).
- `patient_id`
Unique identifier for the patient (optional for modeling, but useful for grouping).

- **age_approx**
Approximate age of the patient at imaging.
- **sex**
Patient's sex (**male/female**).
- **anatom_site_general**
General site of the lesion on the body (e.g. torso, lower extremity).

Other columns may include advanced clinical or image metrics. Refer to the challenge's main documentation for a complete list.

test-metadata.csv

- **isic_id**
Same unique ID structure as in the training data, but **without target**.
- **Other fields** mirror the training set's metadata columns (except **target**).

Submission Format

Your final **submission.csv** must have **two columns**:

1. **isic_id**
The **exact isic_id** from the test set (e.g. "**ISIC_0015670**").
2. **target**
Predicted probability for that lesion being malignant (a float in [0, 1]).

Example:

```
```csv isic_id,target ISIC_0015670,0.80 ISIC_0015701,0.15 ISIC_0015729,0.97 ...
```